DR. KANWAL ASIF

Centre for Endocrinology, William Harvey Research Institute,
Queen Mary University of London (QMUL)
Kanwalkhan338@gmail.com
a.kanwal@qmul.ac.uk

Personal Information
Date of Birth: 22-Mar-1994
Gender: Female
Nationality: Pakistani
Place of Birth: Hyderabad, Sindh

Education

(2019 – 2023) **Doctor of Philosophy (PhD) in Bionanomaterials**
Ca Foscari University of Venice, Italy

(2016 - 2018) **MS Molecular Genetics**
Comsats Institute of Information Technology Islamabad, Pakistan,
CGPA 3.77/4.0

(2012 - 2016) **BS Genetics**
University of Sindh, Jamshoro, Pakistan
CGPA 3.77/4.0

(2009-2011) **Higher Secondary School (Pre-Medical)**
Government College, Hyderabad, Pakistan
Percentage 69%

(2007-2009) **Secondary School Certificate (Pre-Medical)**
Green House School, Hyderabad, Pakistan
Percentage 75%

Career

(11/2023 - present) **Postdoctoral Research Assistant**
Queen Mary University of London

(10/2024 – 12/2024**)** **Visiting Researcher**
Wurzburg Hospital, Germany

(12/2022 – 10/2023) **Research Grant Holder**
Ca Foscari University of Venice, Italy

(9/2019 – 11/2022) **Ph.D. Candidate**
Ca Foscari University of Venice, Italy

(9/2019 – 10/2023) **Researcher**
National Cancer Hospital, CRO, Aviano, Italy

(6/2017 – 6/2018) **Researcher**
Department of Biosciences, COMSATS University, Islamabad Pakistan

(6/2015 – 7/2015) **Internee**
ICCBS, Genomics Department, University of Karachi, Pakistan

(2/2016 – 7/2016) **School Teacher (part time)**
Dawn High School, Hyderabad

Certification / Licence

(12/2023 – 12/2028) UK Home Office Personal Licence (PIL) – In Vivo Research

Awards & Recognitions

(2022) Research grant holder at Ca Foscari University of Venice, Italy

(2022) EACR membership award from European Association Cancer Research.

(2019) Fully funded PhD position at Ca Foscari University of Venice, Italy

(2019) Researcher Award at National Cancer Hospital CRO, Aviano, Italy

(2017) Awarded of the National Internship – Prime Minister Youth Training Scheme

(2017) Awarded merit scholarship – Government Scheme

(2015) Prime Minister Laptop Scheme Award for Undergraduate Program

Oral Presentation and Poster

(2025) **Virtual Oral Presentation** at Joint 24th ENS@T Scientific Meeting, Belgrade, Serbia

(2022) **Oral Presentation** at NanoBio Workshop at Ca'Foscari University, Venice, Italy

(2019-2022) **Oral presentation** for (consecutive three) Annual Bionanomaterials at Ca'Foscari University, Venice, Italy

(2022) **Poster Presentation** in 62nd Annual Meeting of the Italian Cancer Society, Venice, Italy

(2022) **Attended** Black in Cancer Conference, London, UK

(2022) **Attended** EACR Conference Cancer Genomics meeting, Oxford, UK

(2020) **Attended** Immuno Series UK – Virtual

(2019) **Attended** NanoBio Workshop at Ca'Foscari University, Venice, Italy

Invited Talks

(2022) **Oral Presentation** at NanoBio Workshop at Ca Foscari university, Venice, Italy

Committees and Associations:

(2023 - present) Society for Endocrinology

(2023 - present) British Society of Endocrinology

(2022 - 2024) Member of European Association Cancer Research (EACR)

(2022 - 2024) Member of Italian Cancer Society (SIC).

Scientific Skills

- Cell Culture
- 3D organoid culture
- Immuno-oncology
- Gene knockdown using shRNA, CRISPR/CAS9
- Cell and Molecular Biology
- DNA, RNA, Protein samples characterization
- Flow cytometry
- Primer Designing and Optimization
- qPCR Analysis
- Immunohistochemistry
- Immunofluorescence
- Microscopic imaging
- Western Blot
- ELISA
- Cancer mouse model handling to set up (procedures including subcutaneous injection, Intraperitoneal injection, IVIS)
- Data processing
- Statistical analysis
- Programming and Database skills
- Materials synthesis and Characterization

Publications

1- **Kanwal Asif**, Lorenzo Memeo, Stefano Palazzolo, Yahima Frión-Herrera, Salvatore Parisi, Isabella Caligiuri, Vincenzo Canzonieri, Carlotta Granchi, Tiziano Tuccinardi, Flavio Rizzolio STARD3: A Prospective Target for Cancer Therapy, A Review, cancers, *Doi: 10.3390/cancers13184693*

2- **Kanwal Asif**, Muhammad Adeel, Md. Mahbubur Rahman, Isabella Caligiuri, Tiziana Perin, Maja Cemazar, Vincenzo Canzonieri and Flavio Rizzolio Iron Nitroprusside as Chemodynamic Agent and Inducer of Ferroptosis for Ovarian Cancer Therapy, J. Mater. Chem B, *Doi: 10.1039/d2tb02691k*.

3- **Kanwal Asif**, Muhammad Adeel, Md. Mahbubur Rahman, Andrea Augusto Sfriso, Michele Bartoletti, Vincenzo Canzonieri, Flavio Rizzolio, Isabella Caligiuri Silver Nitroprusside as an Efficient Chemodynamic Therapeutic Agent and a Peroxynitrite nanogenerator for Targeted Cancer Therapy, J. Adv Res, *Doi: 10.1016/j.jare.2023.03.005*.

4- **Kanwal Asif**, Muhammad Adeel, Md. Mahbubur Rahman, Michele Bartoletti, Maja Cemazar, Vincenzo, Canzonieri, Flavio Rizzolio, Isabella Caligiuri Copper nitroprusside: An innovative approach for targeted cancer therapy via ROS modulation Biomedicine & Pharmacotherapy, *Doi: 10.1016/j.biopha.2023.116017*.

5- **Kanwal Asif**, Md. Mahbubur Rahman, Andrea Augusto Sfriso, Salvatore Parisi, Vincenzo Canzonieri, Isabella Caligiuri, Flavio Rizzolio, Muhammad Adeel. Biodegradation of Hydroxylated Boron Nitride Nanoplatelets, their Toxic Effect and Drug Delivery Application, *Doi.org/10.1039/D4RA04433A*

6- **Kanwal Asif**, Muhammad Adeel, Md. Mahbubur Rahman, Michele Bartoletti, Maja Cemazar, Vincenzo, Canzonieri, Isabella Caligiuri, Flavio Rizzolio. Self-targeted Nanosystem for enhanced Chemodynamic cancer therapy, *Doi.org/10.1039/D4BM01683A*

7- Muhammad Adeel, Salvatore Parisi, Matteo Mauceri, **Kanwal Asif**, Michele Bartoletti, Fabio Puglisi, Isabella Caligiuri, Md Mahbubur Rahman, Vincenzo Canzonieri, Flavio Rizzolio Self-Therapeutic Nanomaterials for Cancer Therapy: A Review, ACS applied Nano Materials, *Doi: 10.1021/acsanm.0c00762*

8- Muhammad Adeel, **Kanwal Asif**, Md Mahbubur Rahman, Salvatore Daniele, Vincenzo Canzonieri, Flavio Rizzolio Glucose Detection Devices and Methods Based on Metal–Organic Frameworks and Related Materials, *Doi: 10.1002/adfm.202106023*

9- Muhammad Adeel, **Kanwal Asif**, Fahad Alshabouna, Vincenzo Canzonieri, Md Mahbubur Rahman, Sajid Ali Ansari, Firat Güder, Flavio Rizzolio, Salvatore Daniele Label-free electrochemical aptasensor for the detection of SARS-CoV-2 spike protein based on carbon cloth sputtered gold nanoparticles, *Doi: 10.1016/j.biosx.2022.100256*

10- Muhammad Adeel, **Kanwal Asif**, Vincenzo Canzonieri, Hasi Rani Barai, Md Mahbubur Rahman, Salvatore Daniele, Flavio Rizzolio Controlled, partially exfoliated, self-supported functionalized flexible graphitic carbon foil for ultrasensitive detection of SARS-CoV-2 spike protein, *Doi: 10.1016/j.snb.2022.131591*

11- Yaowen Liu; Martin C. Dietl; Robin Heckershoff; Chunyu Han; Hongwei Shi; Matthias Rudolph; Frank Rominger; Isabella Caligiuri; **Kanwal Asif**; Muhammad Adeel; Thomas Scattolin; A. Stephen K. Hashmi Gold-Catalyzed Formal [4+2] Cycloaddition as Access to Antitumor Active Spirocyclic Oxindoles from Alkynes and Isatin-Derived Ketimines, *Doi: 10.1002/anie.202304672*

12- Miriana Di Stefano; Samuele Masoni; Giulia Bononi; Giulio Poli; Salvatore Galati; Francesca Gado; Simone Manzi; Chiara Vagaggini; Annalaura Brai; Isabella Caligiuri; **Kanwal Asif**; Flavio Rizzolio; Marco Macchia; Andrea Chicca; Andrea Sodi; Valeria Di Bussolo; Filippo Minutolo; Philip Meier; Jürg Gertsch; Carlotta Granchi, Elena Dreassi; Tiziano Tuccinardi Design; synthesis. ADME and biological evaluation of benzylpiperidine and benzylpiperazine derivatives as novel reversible monoacylglycerol lipase (MAGL) inhibitors, *Doi: 10.1016/j.ejmech.2023.115916*.

13- Muhammad Adeel, Hong Siog Lee, **Kanwal Asif**, Sabrina Smith, Hasan Kurth, Flavio Rizzolio, Salvatore Daniele, Firat Guder. Micro-Supercapacitors for Self-Powered Biosensors, *Doi: 10.1002/smsc.202400096*

14- Muhammad Adeel, Hong Seok Lee, **Kanwal Asif**, Sabrina Smith, Hasan Kurt, Flavio Rizzolio, Salvatore Daniele, Firat Güder. Microcapacitors for Self-Powered Biosensors, *Doi.org/10.1002/smsc.202400096*

Articles in Submission/preparation:

15- **Kanwal Asif**, Rathi Prasad. Therapeutic Potential of Nanomedicine to Treat Adrenocortical Carcinoma (in preparation).

Patent:

1- M. Adeel, **K. Asif**, F. Rizzolio Bimetallic Metal Nitroprussides for Targeted Cancer Therapy **(International Application No. PCT/IB2023/062285)**

Industrial Innovation /Start-up:

CuRoX Therapeutic Limited (Company number 16274758 *https://findand-update.company-information.service.gov.uk/company/16274758/officers*)

E3 START Course to develop Business:

This program will help me to develop skills to put my Patent in the market and start a business.

Teaching courses:

1- Currently enrolled in PGCAP 2-year program (Queen Mary University of London, London, 2024-2026)
2- Locomotor Module for MBBS 2nd Year Students (PBL (problem-based Based Learning), Queen Mary University of London, London, 2025)
3- Therapeutic Application of Nanomaterials in Cancer Therapy (Ca Foscari University of Venice, Italy, 2021

Supervision/guidance to Student:

1 PhD student at Queen Mary University of London
2 PhD Students at Ca Foscari university of Venice,Italy
3 MS Students at COMSATS University of Islamabad, Pakistan

Google Scholar Information:

Total Citations: 407, h-index: 10, i10-index:11 **(Kanwal Asif)**

Referees:

1- Dr Rathi Prasad (My Postdoc Supervisor)

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Queen Mary University of London (QMUL)
Email: r.prasad@qmul.ac.uk

2- Professor Lou Metherell (Head of Department)

Centre for Endocrinology, William Harvey Research Institute,
Queen Mary University of London (QMUL)
Email: l.a.metherell@qmul.ac.uk

3- Prof. Flavio Rizzolio, (My PhD supervisor)

Department of Molecular Sciences and Nanosystems,
Ca' Foscari University of Venice, Italy.
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